

Development of a 2D Incompressible Navier–Stokes Solver

Finite Difference Formulation with Pressure Poisson Equation

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Abstract

A two-dimensional incompressible Navier–Stokes solver was developed from first principles to strengthen solver-level understanding of computational fluid dynamics beyond commercial workflows. The formulation uses a finite difference discretization on a structured Cartesian grid and a Pressure Poisson Equation (PPE) to enforce incompressibility through pressure–velocity coupling. The solver was applied to the lid-driven cavity problem and validated against the benchmark data of Ghia et al. The results capture the expected recirculating flow structure and show good agreement with reference centerline velocity profiles. This work provided direct insight into discretization, iterative pressure solution, convergence behavior, and the numerical foundations of CFD solver development.

1. Overview

This work was undertaken to build a CFD solver from the governing equations upward and develop a deeper understanding of numerical fluid mechanics beyond the use of pre-built software tools. The lid-driven cavity problem was selected as the validation case due to its well-established benchmark status and its ability to capture key features of incompressible viscous flow, including recirculation, shear-driven motion, and pressure–velocity coupling.

2. Governing Equations

The incompressible continuity and momentum equations are

$$\frac{\partial u}{\partial x} + \frac{\partial v}{\partial y} = 0 \quad (1)$$

$$\frac{\partial u}{\partial t} + u \frac{\partial u}{\partial x} + v \frac{\partial u}{\partial y} = -\frac{1}{\rho} \frac{\partial p}{\partial x} + \nu \left(\frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} \right) \quad (2)$$

$$\frac{\partial v}{\partial t} + u \frac{\partial v}{\partial x} + v \frac{\partial v}{\partial y} = -\frac{1}{\rho} \frac{\partial p}{\partial y} + \nu \left(\frac{\partial^2 v}{\partial x^2} + \frac{\partial^2 v}{\partial y^2} \right) \quad (3)$$

To enforce incompressibility, pressure is obtained from the Pressure Poisson Equation,

$$\nabla^2 p = \frac{\rho}{\Delta t} \left(\frac{\partial u^*}{\partial x} + \frac{\partial v^*}{\partial y} \right) \quad (4)$$

where u^* and v^* are the intermediate velocities prior to pressure correction.

3. Numerical Formulation

The solver was implemented using a structured finite difference framework with the following numerical features:

- second-order central differencing for spatial derivatives,
- explicit time advancement for the momentum equations,
- predictor–corrector treatment for pressure–velocity coupling,
- iterative solution of the PPE using Successive Over-Relaxation (SOR).

The computational sequence consists of:

1. predicting intermediate velocities from the momentum equations,
2. solving the PPE,
3. correcting velocity using the pressure gradients,
4. repeating until convergence is achieved.

The lid-driven cavity was modeled on a square domain with no-slip walls on all boundaries and a constant horizontal lid velocity at the top wall.

4. Key Results

The velocity magnitude contours and streamline field capture the expected structure of the cavity flow, including the dominant primary vortex and the characteristic recirculating motion induced by the moving lid. The solution reproduces the qualitative physics of the benchmark problem and shows stable numerical behavior.

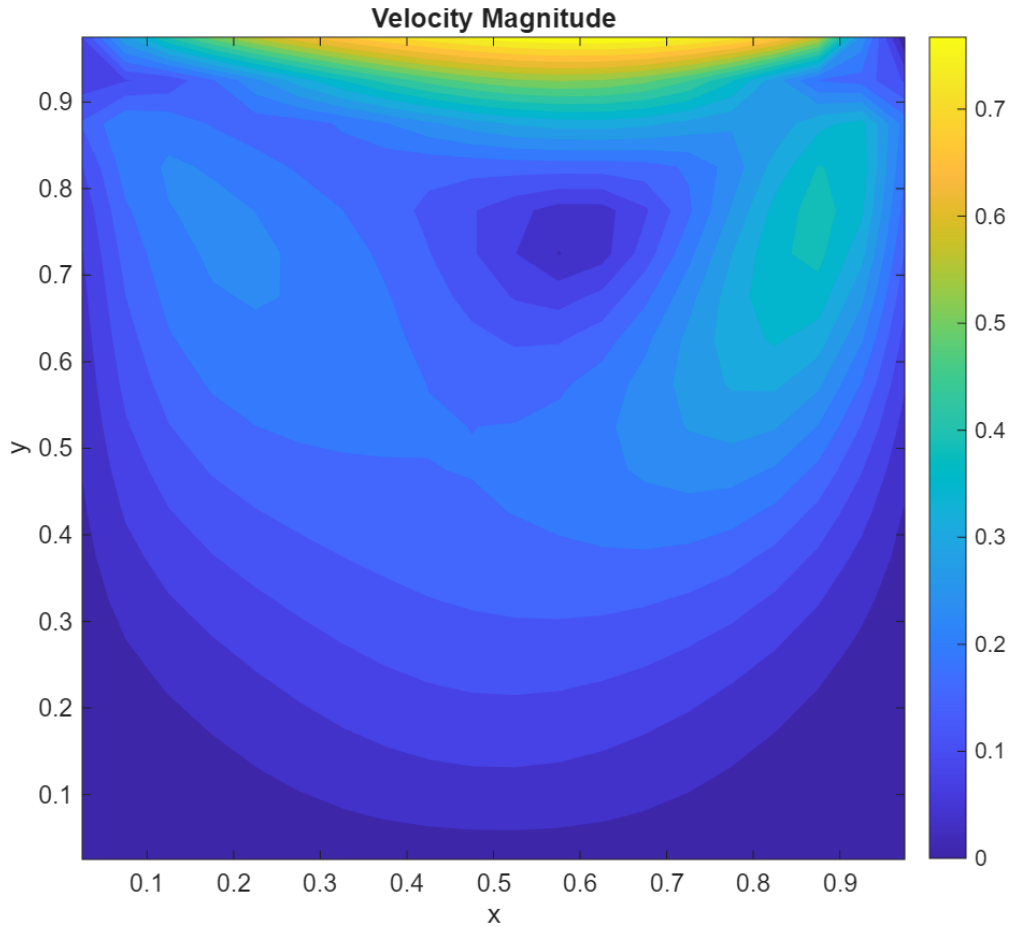


Figure 1: Velocity magnitude contours for the lid-driven cavity flow. The high-velocity region near the moving lid and the progressive decay into the cavity are captured clearly.

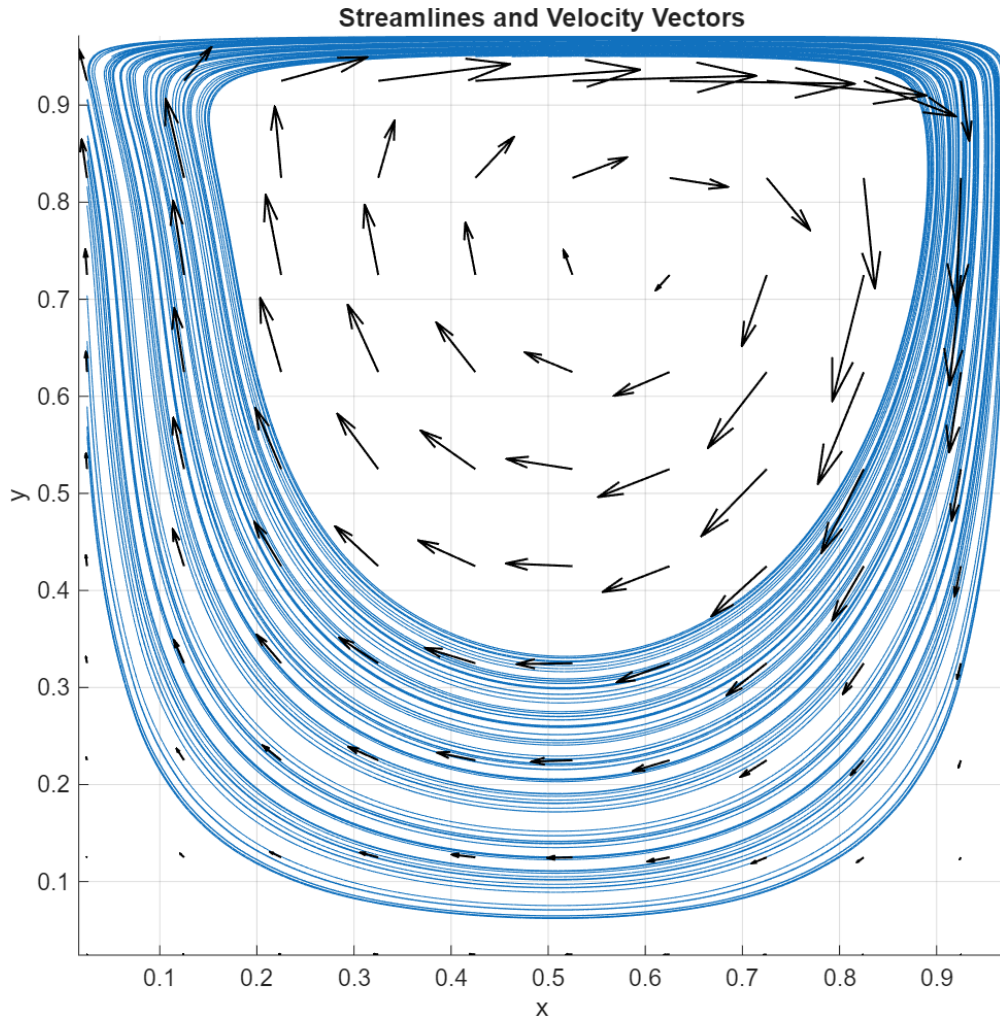


Figure 2: Streamlines and velocity vectors obtained from the present solver. The dominant recirculating structure agrees with the expected topology of incompressible lid-driven cavity flow.

5. Validation

To assess the accuracy of the implementation, the horizontal centerline v -velocity profile was compared with the benchmark data reported by Ghia et al. The present solver shows good agreement with the reference solution and captures the main turning points and extrema of the centerline velocity distribution.

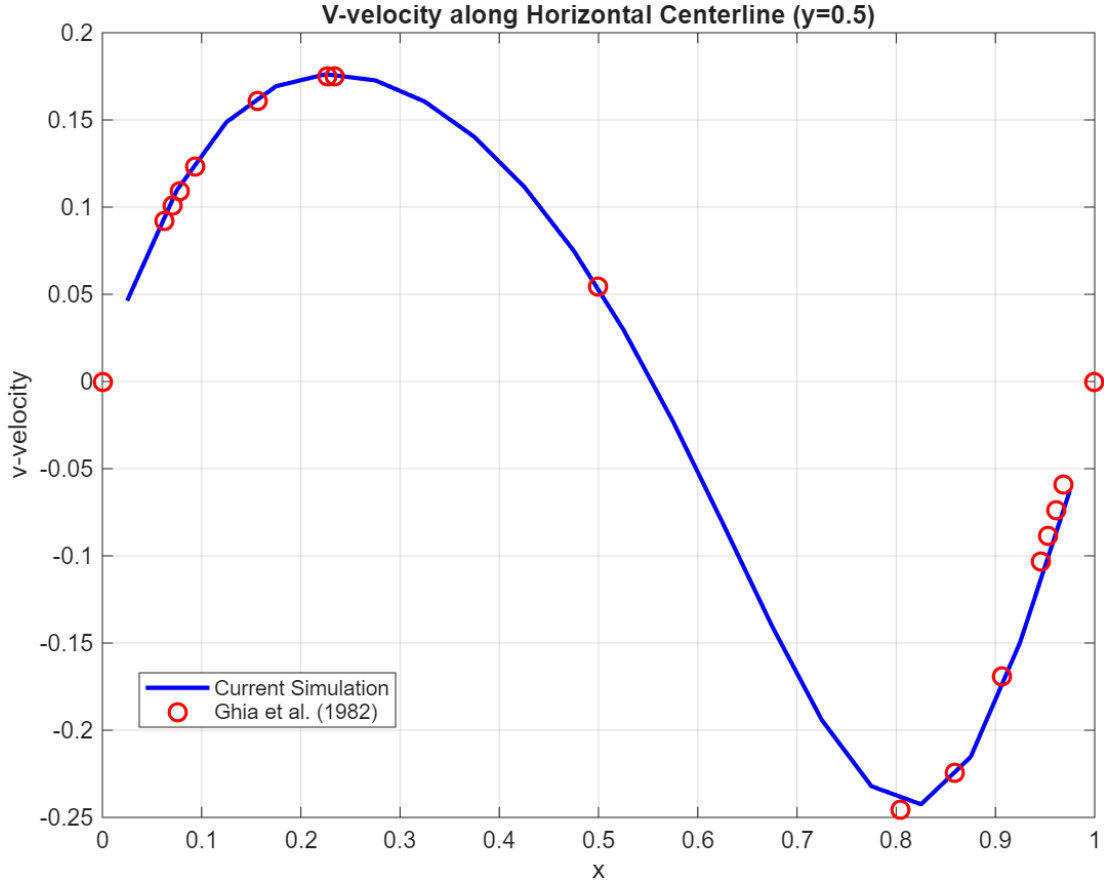


Figure 3: Comparison of the horizontal centerline v -velocity profile with the benchmark data of Ghia et al. The present solver reproduces the overall trend and key extrema of the reference solution.

6. Technical Takeaways

This work provided practical insight into several core CFD concepts:

- pressure–velocity coupling in incompressible flow,
- the role of the Pressure Poisson Equation in enforcing mass conservation,
- sensitivity of convergence to relaxation and time-step selection,
- the influence of discretization and grid resolution on solution quality.

More broadly, the project strengthened my understanding of how governing equations are translated into an operational CFD solver and how numerical choices directly affect stability, accuracy, and physical fidelity.

7. Conclusion

A 2D incompressible Navier–Stokes solver was developed and validated for the lid-driven cavity benchmark using a finite difference formulation with PPE-based pressure correction. The solver reproduced the expected flow physics and showed good agreement with benchmark centerline velocity data. Beyond the numerical results, the project served as a strong foundation for developing solver-level intuition in CFD, particularly in discretization, convergence, and incompressible flow algorithms.

Reference: U. Ghia, K. N. Ghia, and C. T. Shin, “High-Re Solutions for Incompressible Flow Using the Navier–Stokes Equations and a Multigrid Method,” *Journal of Computational Physics*, Vol. 48, No. 3, 1982, pp. 387–411.